Wu 2021 Breast — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

20/20 samples (≥100 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI 1.0→4.96); both lobes (Low-cycling 11,803 / Cycling 12,686) carry near-identical developmental means — no lobe scoping.

Gate 2 — module geography → GO (non-branching)

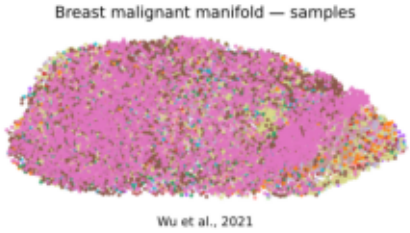
Non-branching signature (spec §1): embryonic programs present with Crest an inverse/terminal gradient — pSom↔Crest r=−0.71, Crest terminal (crest-high decile pt=0.94, rho=+0.55); Gast/pSom body ordered along pseudotime (pt 0.34/0.46), centroids collinear (area/perim²=0.010), cell-cycle-independent (pt↔cyc rho=+0.00). FLAG: Gast/pSom cross-repress (r=−0.47, obs/exp=0.36) — spec-permitted, organized at the body end opposite Crest.

Gate 3 — hub mutual exclusivity → NA

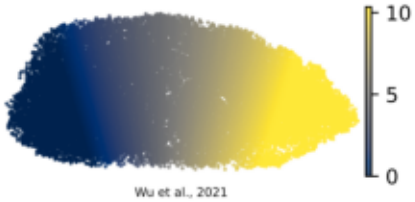
No bifurcation: Gast/pSom repel (obs/exp=0.36) with 0 forks, so the program collapses onto a single axis rather than two arms — there is no branch pair to test for mutual exclusivity.

STEP FIGURES

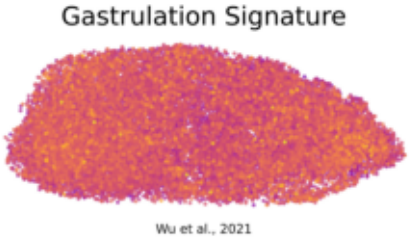
Manifold — samples



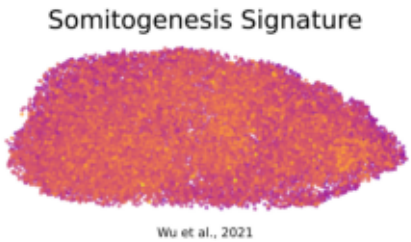
Pseudotime



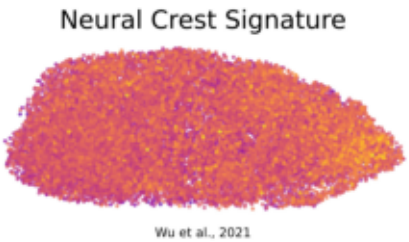
Gastrulation



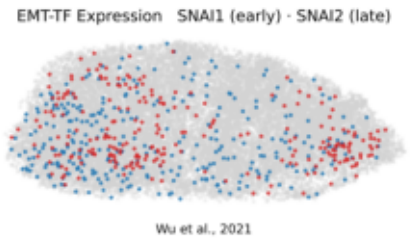
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.

Breast Carcinoma (Wu et al., 2021), 10x — 20 patients / 100,064 cells (master scope); malignant only: 24,489 cells across all 20 samples (≥100 min-cell floor) enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.

Non-branching developmental pattern reproduces — Gate 1 GO, Gate 2 GO (non-branching), Gate 3 NA. Per spec §1–§2, the non-branching case (embryonic programs present, Crest an inverse/terminal gradient, no bifurcation) passes Gate 2. Wu's breast malignant manifold shows exactly that: a Gast/pSom body opposite a terminal Neural-Crest gradient (crest-high decile pt=0.94, rho=+0.55; dominant inverse gradient pSom↔Crest r=−0.71), the programs organized along one axis rather than resolved into two arms. Module-high territories are collinear (area/perim²=0.010) and ordered Gast(pt 0.34)→pSom(pt 0.46) body→Crest terminus, cell-cycle-independent — developmental signal, not proliferation. Atypical feature, flagged: Gast and pSom cross-repress (r=−0.47, obs/exp=0.36) rather than overlap; the spec explicitly permits some cross-repression in the non-branching case, and here the repulsion is organized — at the body end opposite the terminal Crest — not the disorganized spread of a NO-GO. EMT-TFs SNAI1 (root) ↔ SNAI2 (terminus) are near the detection floor (5.2%/4.1%; Δpt=0.09); the axis is carried by the module scores, not a single canonical EMT-TF.

Parameter choices

Min cells/sample	≥100 floor (gap rule ill-defined)	**Integration**	Harmony θ=10; PCA 50→20; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip = Gast/pSom-high, Crest-low	**Compartments**	Low-cycling 11,803 / Cycling 12,686